

Experiment No. : 3

Title: Implement control structures and form validation. Create a grading system based on user-entered marks.

2. Software/Tools Required: Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.

3. Theory:

A. Form Validation

Ensures user input is accurate and valid before processing.

- Empty Check: Ensures the input field is not blank.
- Type Check: Ensures input is a number using `isNaN()`.
- Range Check: Restricts marks within valid bounds.

B. Control Structures (if...else if...else)

Executes specific blocks of code depending on whether conditions evaluate to true or false. Used to map mark ranges to Grades.

C. DOM Manipulation

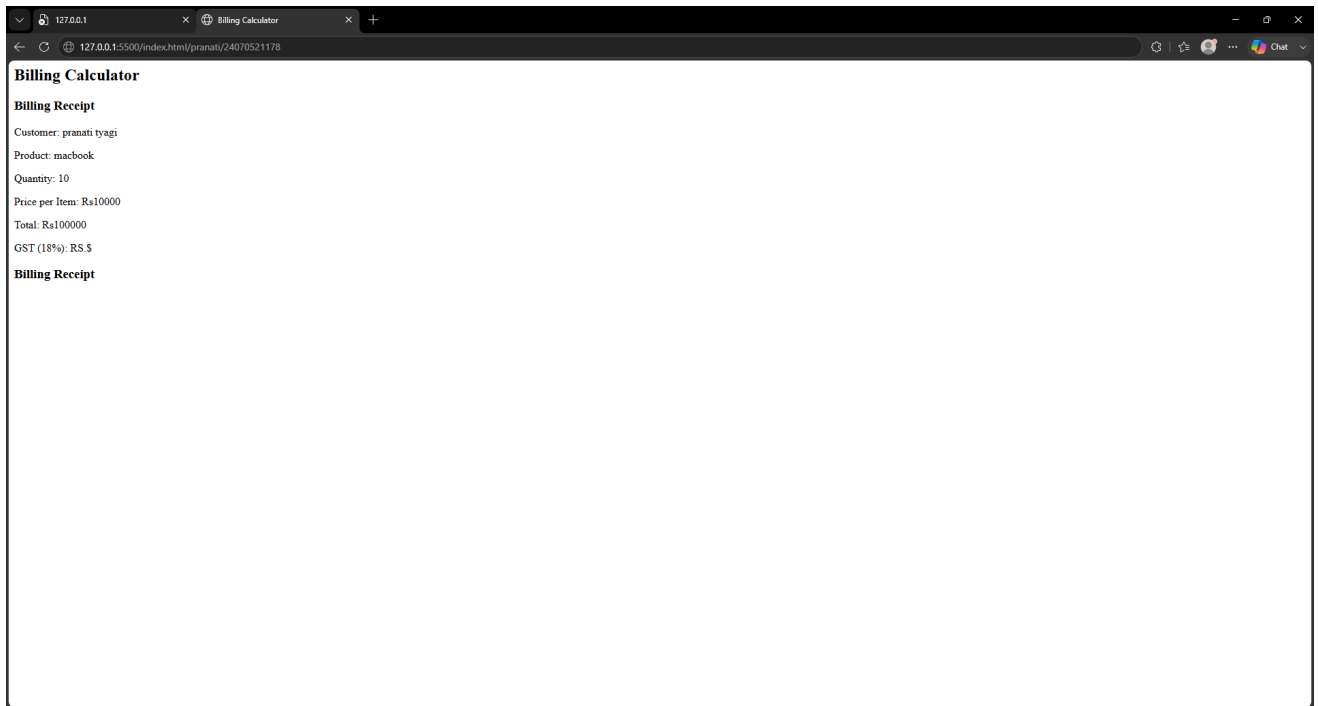
- Input:

`document.getElementById('marks').value` (Reads input).

- Output:

`document.getElementById('result').innerText` (Displays output).

Output:



```

File Edit Selection View Go Run Terminal Help
index.html x scripts.js
<DOCTYPE html> <html> <body> <script> bill
1 <DOCTYPE html>
2 <html>
3 </head>
4 </body>
5
6 <h2>Billing Calculator</h2>
7
8 <script>
9
10 //const example
11 const GST = 0.18;
12 var customerName = prompt("Enter customer name: ");
13 var productname = prompt("Enter product name: ");
14
15 var quantity = Number(prompt("Enter equantity: "));
16
17 let price = Number(prompt("enter price per item:"));
18
19 let total = quantity * price;
20 let Gstamount = total * GST;
21 let finalbill = total + Gstamount;
22
23 const bill = {
24   customer: customerName,
25   product: productname,
26   quantity,
27   price,
28   total,
29   Gstamount,
30   finalbill
31 };
32
33 const {
34   customer,
35   product,
36   Gstamount: gst,
37   finalbill: amount
38 } = bill;
39
40 document.write(<h3>Billing Receipt</h3>);
41 document.write(<p>Customer: ${customer}</p>);
42 document.write(<p>Product: ${product}</p>);
43 document.write(<p>Quantity: ${quantity}</p>);
44 document.write(<p>Price per Item: Rs${price}</p>);
45 document.write(<p>Total: Rs${total}</p>);
46 document.write(<p>GST (18%): RS.${gst}</p>);
47 document.write(<h3>Billing Receipt</h3>);
48
49 </script>
50
Ln 26, Col 34 Spaces: 4 UTF-8 CRLF HTML Port: 5500

```

5. Result/Conclusion: By using control structures (if...else if) alongside form validation techniques, web applications can accurately process user inputs, prevent faulty data processing, and produce dynamic, real-time results such as student grading reports.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

5. Case Study Title:

Task 1: Scholarship Eligibility

Case Study:

A college awards scholarships based on marks.

Below 50 - Not Eligible

50–69 - Bronze

70–84 - Silver

85–94 - Gold

95–100 - Platinum

Task:

Write a JavaScript program that accepts student marks and displays the scholarship category.

6. Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Scholarship Eligibility</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f2f2f2;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }

    .container {
      background: white;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 0 10px rgba(0,0,0,0.2);
      width: 350px;
      text-align: center;
    }

    h2 {
      color: #2c3e50;
    }
  </style>
</head>
<body>
  <div class="container">
    <h2>Scholarship Eligibility</h2>
    <div>
      <input type="text" value="Enter Marks" />
      <input type="button" value="Check" />
    </div>
    <div>
      <div>Below 50 - Not Eligible</div>
      <div>50–69 - Bronze</div>
      <div>70–84 - Silver</div>
      <div>85–94 - Gold</div>
      <div>95–100 - Platinum</div>
    </div>
  </div>
</body>
</html>
```



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

```
input {
  width: 90%;
  padding: 10px;
  margin: 10px 0;
  font-size: 16px;
}

button {
  padding: 10px 20px;
  background: green;
  color: white;
  border: none;
  cursor: pointer;
  font-size: 16px;
  border-radius: 5px;
}

button:hover {
  background: darkgreen;
}

#result {
  margin-top: 20px;
  font-size: 18px;
  font-weight: bold;
  color: blue;
}
</style>
</head>
<body>

<div class="container">
  <h2>Scholarship Eligibility</h2>

  <label>Enter Student Marks:</label><br>
  <input type="number" id="marks" placeholder="Enter marks (0-100)">

  <br>
  <button onclick="checkScholarship()">Check</button>

  <p id="result"></p>
</div>

<script>
function checkScholarship() {

  let marks = Number(document.getElementById("marks").value);
```




SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

8.Result/Conclusion: The grading system successfully accepted the user's marks, validated the input, and displayed the correct grade according to the predefined grading criteria. This experiment helped in understanding how form validation ensures valid user input and how control structures (if-else statements) are used to make decisions and generate appropriate output in a web application.